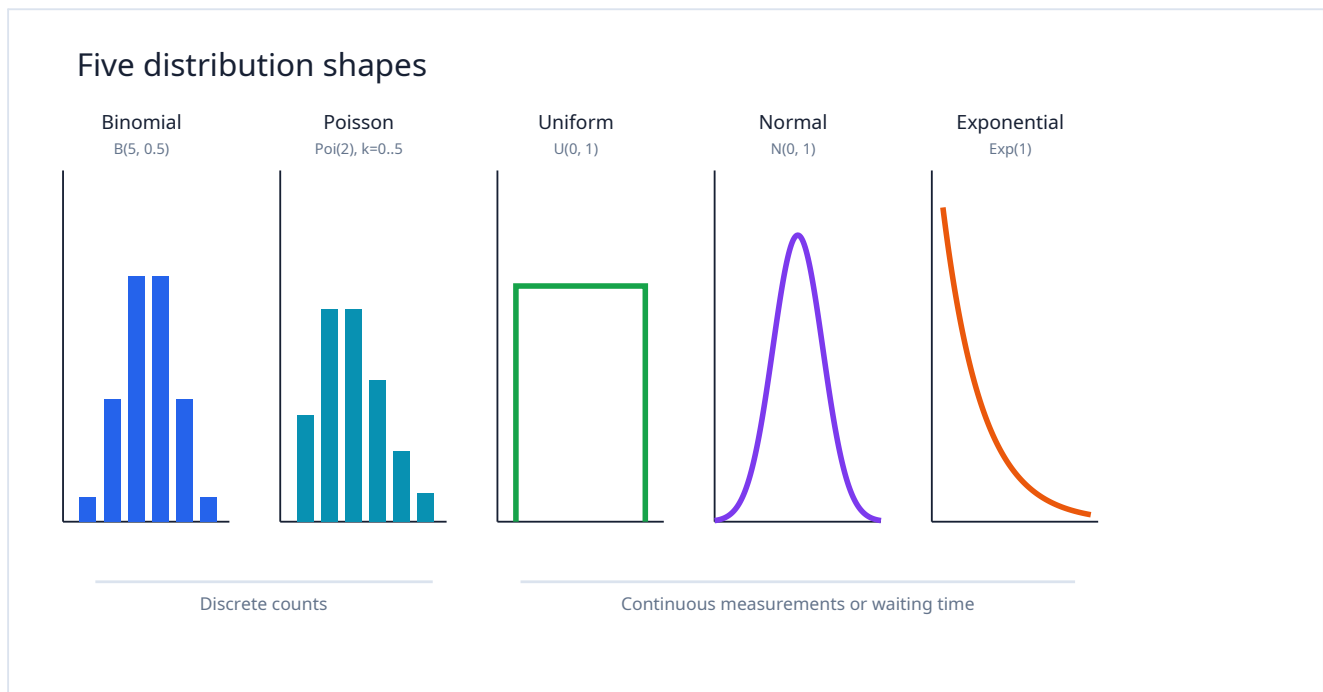
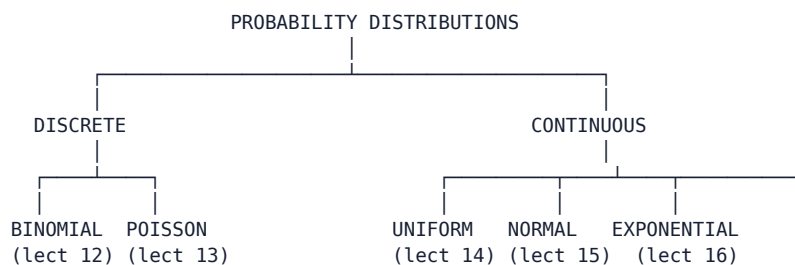


07 UNIFORM, NORMAL, EXPONENTIAL (ppt4, 44 pages, lectures 14-16)

The family tree first (p001)



Five distribution shapes for comparison



PART A: UNIFORM (p002-p006) - lecture 14

1. Definition

constant density, so equal-length subintervals have equal probabilities. Individual points each have probability zero.

$$f(x) = 1/(b-a) \quad a \leq x \leq b$$

$$f(x) = 0 \quad \text{otherwise}$$

the flat rectangle:



the area is width $\times$ height $= (b-a) \times 1/(b-a) = 1$ ✓

2. Properties

$$\mu = (a + b)/2$$

$$\sigma^2 = (b - a)^2/12 \quad (\text{the 12 is FIXED})$$

THE LENGTH-RATIO SHORTCUT: probability = overlap length divided by total length. No integration needed.
 $P(c \leq X \leq d) = \text{length}([c,d] \text{ intersection } [a,b]) / (b-a)$
 If $[c,d]$ lies inside $[a,b]$, this becomes $(d-c)/(b-a)$.

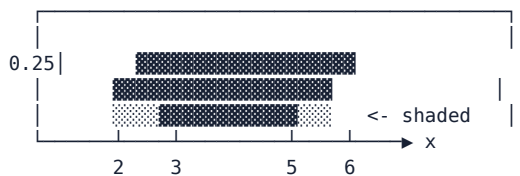
3. WORKED: U(2,6) (p004-p006)

$$f(x) = 1/(6-2) = 0.25, \quad 2 \leq x \leq 6$$

$$\mu = (2+6)/2 = 4$$

$$\sigma^2 = (6-2)^2/12 = 16/12 = 1.333$$

$$P(3 < X < 5) = \text{length ratio} = (5-3)/(6-2) = 2/4 = 0.50$$



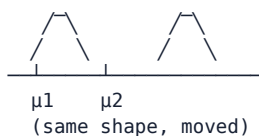
$$P = \text{shaded} / \text{total} = 2/4 = 0.50$$

PART B: NORMAL (p007-p037) - lecture 15

4. Shape facts (p008-p010)

BELL SHAPED
 SYMMETRICAL
 MEAN = MEDIAN = MODE = μ
 range: $-\infty$ to $+\infty$
 location set by μ , spread set by σ

changing μ shifts:



changing σ widens:



5. The notation and the density (p010-p011)

$$X \sim N(\mu, \sigma^2)$$

THE SECOND SLOT IS THE ****VARIANCE****, not the SD
 "mean 70, variance 25" $\rightarrow \mu=70, \sigma^2=25, \sigma=5$
 ALWAYS square-root it before a z-score.

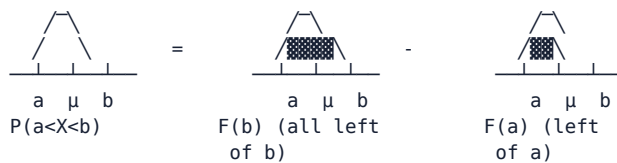
$$f(x) = 1/(\sigma\sqrt{2\pi}) \cdot e^{-\{(x-\mu)^2/(2\sigma^2)\}}$$

6. The cdf and interval (p012-p013)

$$F(x_0) = P(X \leq x_0) \quad (\text{area to the left})$$

$$P(a < X < b) = F(b) - F(a)$$

the subtraction picture:

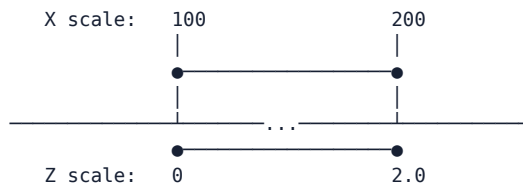


7. Standardisation (p014-p015)

$$Z = (X - \mu) / \sigma \quad Z \sim N(0, 1)$$

WORKED: $\mu=100$, $\sigma=50$, $X=200$
 $Z = (200-100)/50 = 2.0$

the two scales, same shape:



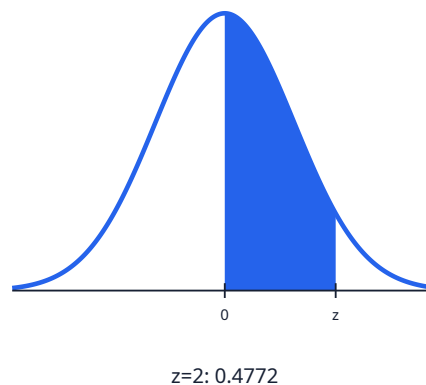
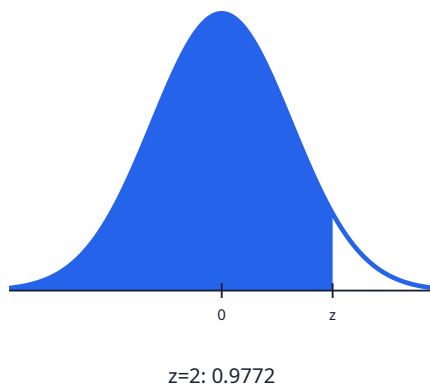
"the distribution is the same, only the SCALE has changed"

8. THE TWO TABLES - the biggest trap in the whole paper (p022-p023)

Two normal-table conventions

p022: cumulative $P(Z < z)$

p023: centre area $P(0 < Z < z)$



Cumulative and centre-area normal tables

THE SLIDES SHOW **TWO DIFFERENT TABLES** AND THEY LOOK ALIKE

TABLE 1 (p022, cumulative standard normal table):
 gives $P(Z < z)$ = CUMULATIVE from $-\infty$ to z
 $F(z)$. e.g. $z=0.0 \rightarrow 0.5000$, $z=2.0 \rightarrow 0.9772$

TABLE 2 (p023, area from 0 to z):
 gives $P(0 < Z < z)$ = area FROM THE CENTRE to z
 $A(z)$. e.g. $z=2.0 \rightarrow 0.4772$

THEY DIFFER BY 0.5 FOR $z > 0$:

$$F(z) = 0.5 + A(z)$$

$$0.9772 = 0.5 + 0.4772 \quad \checkmark$$

DECIDE WHICH TABLE THE QUESTION GIVES **BEFORE** SUBSTITUTING

the two areas, drawn:

$$\phi(z) = P(0 < Z < z)$$



0 z
centre to z

$$F(z) = P(Z < z)$$



$-\infty$ z
everything to z

9. Symmetry facts (p024-p025)

$$P(Z < -2.00) = 1 - P(Z < 2.00) = 1 - 0.9772 = 0.0228$$

$$\Phi(-z) = 1 - \Phi(z)$$

$$P(Z > z) = 1 - \Phi(z)$$

the mirror picture:



-2.0 0
left tail .0228



0 2.0
right tail .0228

the left tail at -z EQUALS the right tail at +z

10. WORKED: the 8.6 chain (p028-p033)

$$X \sim N(8.0, 5.0^2) \quad [\text{note: } \sigma = 5.0]$$

(a) FIND $P(X < 8.6)$:

$$Z = (8.6 - 8.0)/5.0 = 0.12$$

$$P(X < 8.6) = P(Z < 0.12) = F(0.12) = 0.5478$$

(b) FIND $P(X > 8.6)$:

$$P(X > 8.6) = 1 - 0.5478 = 0.4522$$

the two areas:



8.0 8.6
 $P(X < 8.6) = .5478$



8.0 8.6
 $P(X > 8.6) = .4522$

ERRATA 8: the figure on p030 labels the curve " $\sigma = 10$ ", but the arithmetic on the same page divides by 5.0. Use $\sigma = 5.0$.

11. WORKED: the inverse, 20% lower tail (p034-p037)

Find x so that only 20% of values are BELOW x.

STEP 1: find z for a 20% lower tail

look for $F(z) = 0.20$

from the table, $z \approx -0.84$ (exactly -0.8416)

STEP 2: CONVERT BACK TO X ← the forgotten step

$$X = \mu + Z\sigma = 8.0 + (-0.84)(5.0) = 8.0 - 4.2 = 3.80$$

THE INVERSE HAS TWO STEPS AND PEOPLE STOP AFTER STEP 1.

Getting $z = -0.84$ is HALF the answer.

The mark is in $X = \mu + z\sigma$.

the picture:

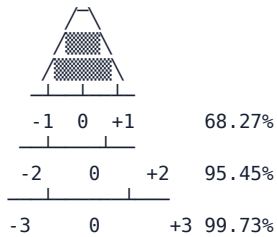


3.80 8.0
.20 .80

▲
 $z = -0.84$

12. The landmarks (the hidden item H4)

within 1σ : 68.27%
 within 2σ : 95.45%
 within 3σ : 99.73%



PART C: EXPONENTIAL (p038-p043) - lecture 16

13. What it models

the LENGTH OF TIME between two occurrences of an event
 (the time between arrivals)

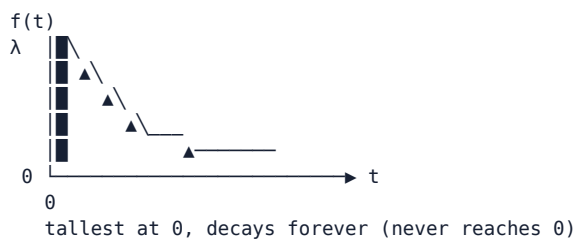
examples: trucks at a dock, transactions at an ATM, phone calls

14. The density and its parameters (p040-p041)

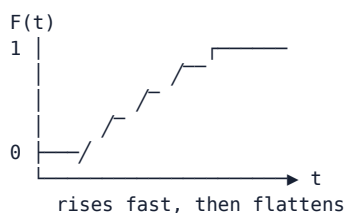
$$\begin{aligned} f(t) &= \lambda e^{-\lambda t}, & t > 0 \\ \mu &= 1/\lambda & \sigma^2 &= 1/\lambda^2 \\ F(t) &= 0 & \text{for } t \leq 0 \\ &= 1 - e^{-\lambda t} & \text{for } t > 0 \\ P(T > t) &= 1 & \text{for } t < 0 \\ &= e^{-\lambda t} & \text{for } t \geq 0 \end{aligned}$$

ERRATA 4: the slide on p040 says "1/λ is the MEAN NUMBER of occurrences per unit time". THAT IS BACKWARDS.
 λ = the rate (occurrences per unit time)
 $1/\lambda$ = the mean TIME between occurrences
 p041 states it correctly as $\mu = 1/\lambda$.

THE DECAY PICTURE:



the cdf rises toward 1:



15. WORKED: arrivals at 15 per hour (p042-p043)

"customers arrive at the rate of 15 per hour.
 $P(\text{the arrival time between consecutive customers} < 3 \text{ minutes})$?"

$\lambda = 15$ per HOUR

```

3 minutes = 0.05 hours      <- CONVERT THE UNIT
P(T < 0.05) = 1 - e^{-λt}
              = 1 - e^{-(15)(0.05)}
              = 1 - e^{-0.75}
              = 1 - 0.4724 = 0.5276

```

THE UNIT CONVERSION IS THE WHOLE QUESTION.
 λ is per hour, so t MUST be in hours.
 3 min = 0.05 hr.
 This is the trap the setter builds into almost every exponential story (errata families E24S4, A2 D2).

the mean gap = $1/\lambda = 1/15$ hour = 4 minutes

16. The memoryless property (hidden item H3)

$$P(T > s + t \mid T > s) = P(T > t)$$

"surviving s MORE, given you already survived to now, is the same as surviving s from scratch"
 the exponential NEVER AGES. A used part is as good as new. (unlike, say, a lightbulb that wears out)

PROOF SKETCH:

$$\begin{aligned}
 P(T > s+t \mid T > s) &= P(T > s+t \text{ and } T > s) / P(T > s) \\
 &= P(T > s+t) / P(T > s) \\
 &= e^{-\lambda(s+t)} / e^{-\lambda s} \\
 &= e^{-\lambda t} = P(T > t) \quad \checkmark
 \end{aligned}$$

the $e^{-\lambda s}$ cancels <- the whole trick

the canonical number: $e^{-1} = 0.3679$

THE THREE CONTINUOUS DISTRIBUTIONS, ONE TABLE

	UNIFORM	NORMAL	EXPONENTIAL
pdf	$1/(b-a)$	$1/(\sigma\sqrt{2\pi})e^{..}$	$\lambda e^{-\lambda t}$
mean	$(a+b)/2$	μ	$1/\lambda$
variance	$(b-a)^2/12$	σ^2	$1/\lambda^2$
cdf	0, ratio, 1	table	0 or $1-e^{-\lambda t}$
tail	ratio	$1-\Phi(z)$	$e^{-\lambda t}$
parameter	a, b	μ, σ^2	λ (a RATE)
models	random pt	measurement	waiting time

the three shapes side by side:

